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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026 - Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCMBG301 – ESSENTIALS OF IMMUNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes Maximum Marks: 50

Part A - Short Note Questions - Answer any 10 questions - Each question carries 2 marks

1. Which organism is used to produce the smallpox vaccine? [K] / [CO1]

Vaccinia virus [2]

2. What is the antigen-binding site of an immunoglobulin? [K] / [CO2]

Paratope [2]

3. What is an Arthus reaction? [K] / [CO3]

Local Type III hypersensitivity reaction caused by immune complex deposition [1 + 1]

4. What is the acute phase response? [K] / [CO1]

Systemic response to infection or inflammation [1]; increased production of acute-phase proteins / CRP/ MBL [1]

5. Give one example of a tube flocculation test. [K] / [CO2]

VDRL test [2]

6. What is the role of the thymus in the immune system? [K] / [CO1]

Site of T-lymphocyte maturation [2]

7. What is the purpose of using HAT medium in hybridoma technology? [U] / [CO2]

Selects hybridoma cells [1]; allows hybrid cells to survive while unfused cells die [1]

8. What is the role of antibodies in ADCC? [An] / [CO3]

Antibodies bind to target cells [1]; NK cells recognise the Fc portion, bind and kill target cells [1]

9. What is an endotoxin? [K] / [CO1]

Lipopolysaccharide (LPS) [1]; present in the outer membrane of Gram-negative bacteria [1]

10. Define the complement system. [K] / [CO2]

Group of plasma proteins [1]; helps in immune defence through inflammation, opsonisation and cell lysis [1]

11. **Name the two types of agranulocytes. [K] / [CO1]**

Monocytes [1]; lymphocytes [1]

12. **Name two systemic autoimmune diseases and mention their major target tissues.[K][CO3]**

Systemic lupus erythematosus (SLE) – multiple tissues, including skin, joints and kidneys [1]; rheumatoid arthritis – joints [1]; or Multiple sclerosis – myelin sheath

[2x10 = 20]

Part B - Short Essay Questions - Answer any 4 questions - Each question carries 5 marks

13. List the secondary lymphoid organs and summarise their functions in the immune response. [K] / [CO1]

- Lymph nodes [1] - Spleen [1] - Mucosa-associated lymphoid tissue (MALT) [1] - Trap and filter antigens [1] - Provide sites for lymphocyte activation and immune responses [1]

14. Compare RIST and RAST based on their purpose and detection. [An] / [CO3]

- RIST detects total serum IgE [1] - RAST detects allergen-specific IgE [1] - Both are used in allergy diagnosis [1]- RIST measures total IgE concentration [1]- RAST identifies IgE against a specific allergen [1]

15. Explain how activated B cells differentiate into antibody-secreting plasma cells.[U] / [CO3]

- B cells recognise specific antigens [1] - Antigen activates B cells, usually with helper T-cell support [1] - Activated B cells undergo clonal proliferation [1]- B cells differentiate into plasma cells [1] - Plasma cells secrete specific antibodies [1]

16. Explain the steps involved in indirect ELISA. [U] / [CO2]

- Antigen is coated onto the microtitre plate [1] - Patient serum is added [1] - Enzyme-linked secondary antibody is added [1] - Substrate is added [1] - Colour development indicates the presence of specific antibodies [1]

17. What are the major features of acquired immunity? [An] / [CO1]

- Antigen specificity [1] - Diversity [1]- Immunological memory [1]- Self/non-self discrimination [1] - Clonal expansion of specific lymphocytes [1]

18. Describe the two stages involved in the complement fixation test. [K] / [CO2]

- Stage 1: Patient serum is mixed with known antigen and complement [1.5] - If specific antibodies are present, antigen–antibody complexes fix the complement [1] - Stage 2: Sensitised sheep red blood cells are added as an indicator system [1.5] - No haemolysis indicates a positive test; haemolysis indicates a negative test [1]

[5x4 = 20]

Part C - Essay Question - Answer any 1 question - Each question carries 10 marks

19. What are adjuvants, and explain the mechanism of immune stimulation by adjuvants. [U] / [CO2]

- Definition of adjuvants [2]. Examples: Alum, Freund's adjuvant [2]. Enhance antigen uptake by antigen-presenting cells [2]. Stimulate innate immune responses and cytokine production [2]. Improve lymphocyte activation and antibody production [2]

20. Explain the primary and secondary immune responses with suitable diagrams. [U] / [CO3]

- Definition of primary immune response [2]. Definition of secondary immune response [2]. Primary response: slow, longer lag period and lower antibody level [2]. Secondary response: rapid, higher antibody level and longer duration [2]. Suitable labelled diagrams showing antibody response over time [2]

[10x1 = 10]